

Alejandro Rojas Bernal

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Contact Information

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Fields

Macroeconomics, International Finance, International Economics, Production Networks

Education

Ph.D. in Economics - University of British Columbia <i>Emphasis in macroeconomics and international finance</i>	Vancouver, Canada 2024
Master of Arts in Economics - Universidad de Los Andes <i>Emphasis on finance and econometrics</i>	Bogotá, Colombia 2018
Bachelor of Arts in Law - Universidad de los Andes <i>Summa Cum Laude</i>	Bogotá, Colombia 2016
Bachelor of Arts in Economics - Universidad de Los Andes <i>Cum Laude</i>	Bogotá Colombia 2016

Publications

1. [Pricing the exotic: Path-dependent American options with stochastic barriers](#)

with [Mauricio Villamizar-Villegas](#)

Latin American Journal of Central Banking

We develop a novel pricing strategy that approximates the value of an American option with exotic features through a portfolio of European options with different maturities. Among our findings, we show that: (i) our model is numerically robust in pricing plain vanilla American options; (ii) the model matches observed bids and premiums of multidimensional options that integrate ratchet, Asian, and barrier characteristics; and (iii) our closed-form approximation allows for an analytical solution of the option's greeks, which characterize the sensitivity to various risk factors. Finally, compared to the traditional Monte Carlo simulations method, we highlight that our estimation's prediction is more accurate and requires less than 1% of the computational time.

Working Papers

1. **Inequality and Misallocation under Production Networks**

In this paper, I develop an aggregation theory for distorted production network economies with heterogeneous households. I provide general decompositions for how the aggregate and distributional effects of shocks are sensitive to underlying consumer and firm heterogeneity. The workers' value-added over labor income ratios (distortion centralities) gauge the importance of workers in the production of heavily distorted firms and are sufficient statistics for the effect of income distribution variations on TFP. The average distortion centrality faced by a household's expenditure (expenditure centrality) and a firm's revenue (revenue centrality) are sufficient statistics for the effect of expenditure variations on TFP. Labor misallocation rises and TFP falls as labor income shifts toward high distortion centrality workers, consumption shifts toward high expenditure centrality households, or demand shifts toward high revenue centrality firms. The reason is that when aggregate expenditure on relatively undistorted firms rises, their labor demand increases, reallocating workers from distorted firms with high marginal productivity to relatively undistorted firms with low marginal productivity. These second-best results show how distributional variations affect aggregate output by changing the aggregate allocation efficiency of workers. I estimate the first production network model with household heterogeneity for the United States. I show that variations in the income distribution have been responsible for 20% of the TFP volatility. Additionally, income distribution variations reduced misallocation between 2001 and 2009, and accentuated misallocation after the Great Recession. Heterogeneities in the production network are essential in explaining income and real consumption inequalities.

2. **International Misallocation and Comovement under Production Networks**

In this paper, I develop a general aggregation theory that explains the role of production networks in country-level TFP. This theory applies to a distorted production network open economy with endogenous factor supply. My main contribution is to provide decompositions for the country-level TFP variation that account for the possibility that factors of production and dividends cross national boundaries. The country-level TFP depends on sufficient statistics that characterize the effect on domestic real GDP from (i) firm-level productivity and markdown shocks in domestic and foreign firms and (ii) variations in the global income distribution. These decompositions do not require quantity measures of variations, facilitating their empirical implementation, as price data is no longer necessary. Additionally, for an efficient economy, a Hulten type of decomposition exists for each country, and the global sales distribution is a sufficient statistic to characterize the first-order propagation of global shocks on country-level TFP. These results support a theory of economic spillovers and contagion through industrial networks, corroborating the essential role of global value chains in creating strong complementarities and commonalities in business cycles across countries.

3. **Growth Through Industrial Linkages**

with Xiaojun Guan

In this paper, we estimate how changes in the structure of global input-output networks have influenced growth. Using an open economy production network model, we identify sufficient statistics that characterize how productivity shocks across domestic and foreign firms influence country-level TFP. We estimate these sufficient statistics using data on input-output networks and sectoral productivity shocks. Structural changes in global input-output networks between 1965 and 2000 were advantageous

for developing countries and unfavorable for advanced economies. Holding the global input-output network fixed, TFP growth in China and India would have been 26.6% and 9.7% lower between 1965 and 2000. Whereas for the US and Australia, TFP growth would have been 4.0% and 16.8% higher. Finally, we show that the dynamics of the domestic intermediate input cost share capture the importance that the structure of the global input-output network has on the amplification of shocks on TFP. Our analysis illustrates the importance of industrial linkages and robust domestic intermediate input markets for economic growth.

4. **Nonlinearities in Production Network Economies with Distortions**

with **Ali Karimirad**

We offer the first non-linear analysis of how microeconomic disruptions in an inefficient production network economy impact aggregate TFP. Our decompositions, applicable to any general equilibrium economy, provide non-parametric insights. We identify essential general equilibrium metrics for capturing the non-linear repercussions of microeconomic fluctuations through network connections. Our findings encompass firm-level productivity shocks, wasted distortions, and rebated distortions. We reveal that substantial shocks or high production process complementarity/substitution introduce substantial bias in the linear approximation of distortions and productivity shocks found in the literature. Our non-linear second-order effects substantially mitigate this bias.

5. **Stairway To Haven**

This paper attempts to identify the main channels for the propagation of the macroeconomic effects from corporate profit shifting into tax havens. This question is answered by building a general equilibrium model that introduces firm profit shifting to tax havens in a multi-country environment with production networks. In this model, haven jurisdictions specialize and compete for shifted profits by trading concealment assets in a differentiated oligopolistic environment, and non-haven countries defend these profits by setting enforcement levels over capital flows. The central point of the model is that profit shifting introduces two classes of optimal distortions, first, rebated distortions that by modifying the terms of trade and the effective marginal tax rate alter the decision of firms, but also wasted distortions that optimally squander resources via enforcement policies and the corporate costs that firms have to incur in order to access and develop concealment strategies. I show that the main transmission channels for the propagation of these distortions occurs by increasing corporate dividends, the tax base, and wages in tax havens; while non-haven countries are affected by opposite effects in addition to the wasted distortions. We confirm these results in a three country one sector global economy that additionally provides evidence about the relevance of the structure of the production network and the consumption bundle in the magnitude of the effect from introducing profit shifting.

Presentations

2025

- 10th Annual International Conference in Hawai'i organized by Kobe University
- 2025 World Congress of the Econometric Society, Seoul, Korea
- ECINEQ 2025, Society for the Study of Economic Inequality
- Western Economic Association 100th Annual Conference, San Francisco, California
- Midwest Macro Meeting, Federal Reserve Bank of Kansas, Kansas City

- 32nd Symposium Society for Nonlinear Dynamics and Econometrics, San Antonio, Texas

2024

- American Social Science Association Meeting, San Antonio, Texas
- Bank of Canada, Ottawa, Ontario
- Hawai'i-Hitotsubashi-Keio (H2K) Workshop on International Economics, University of Hawai'i at Manoa, Honolulu, Hawai'i
- University of Hawai'i at Manoa, Honolulu, Hawai'i
- Sao Paulo School of Economics, Sao Paulo, Brazil
- Tecnologico de Monterrey, Mexico D.F., Mexico
- Tilburg University, Tilburg, Netherlands

2023

- North American Summer Meeting of the Econometric Society Meeting, UCLA, Los Angeles
- Asian Meeting of the Econometric Society, East and Southeast Asia, NTU, Singapore
- Australasian Meeting of the Econometric Society, UNSW, Sydney
- Canadian Economic Association, U-Manitoba and U-Winnipeg, Winnipeg

2021

- Quantil: Applied Mathematics Seminar, Bogotá

Teaching Experience

University of Hawai'i at Manoa

- **Principles of Macroeconomics**, Econ 131 (2025, 2026)
- **Global Economic Crisis and Recovery**, Econ 414 (2024, 2025)
- **International Macroeconomics (Master and PhD)**, Econ 662 (2025)

University of British Columbia - Teaching Assistant

- **International Finance (Master and PhD)**, Econ 556, Michael B. Devereux (2023)
- **International Macroeconomics and Finance**, Econ 456, Michael B. Devereux (2022, 2023)
- **Monetary Theory**, Econ 447, Geoffrey Newman (2020, 2021, 2022)
- **International Finance**, Econ 356, Geoffrey Newman (2019, 2020, 2021, 2022)
- **Money and Banking**, Econ 345, Geoffrey Newman (2023)
- **Law and Economics**, Econ 367, Sam Norris (2021)

Universidad de Los Andes - Teaching Assistant

- **Advanced Econometrics (Master and PhD)**, Raquel Bernal and Diego Amador (2017, 2018)
- **Introduction to Macroeconomics**, Marc Hofstetter (2017, 2018)

Universidad de Los Andes - Undergraduate Teaching Assistant

- **Financial Derivatives**, Jairo Rendon, Financial Derivatives (2016)
- **Intermediate Econometrics**, Jorge Alexander Bonilla (2015, 2016)
- **Constitutional Hermeneutics**, Daniel Eduardo Bonilla (2014)
- **General Criminal Law**, Ricardo Posada (2014)
- **Special Criminal Law**, Ricardo Posada (2013)

- **General Criminal Law**, Yamila Fakhouri (2013)
- **Constitutional Law**, Diego Eduardo López (2011)
- **Roman Law**, Nicolás Pájaro (2011)

Professional Experience

- **University of Hawai'i at Manoa**: Assistant Professor, Department of Economics (2024 - Ongoing)
- **Central Bank of Colombia**: Analyst in Financial Markets Development (2016 - 2018)

Scholarships Awards and Honors

2025

- College of Social Sciences Travel Award, University of Hawai'i at Manoa: \$1,425

2023

- International Tuition Award UBC: \$2,633
- Graduate Student Travel Award UBC: \$500
- President's Academic Excellence Initiative PhD Award UBC: \$475

2022

- Faculty of Arts Graduate Award UBC: \$7,511
- Bank of Montreal Graduate Fellowship: \$4,583
- International Tuition Award UBC: \$3,199
- President's Academic Excellence Initiative PhD Award UBC: \$1,620

2021

- Bank of Montreal Graduate Fellowship: \$9,166
- International Tuition Award: \$3,199
- Kathleen Simpson Memorial Graduate Scholarship: \$2,250
- President's Academic Excellence Initiative PhD Award UBC: \$515
- Honorable Mention in the Central Bank Award Rodrigo Gómez CEMLA

2020

- AD Scott Fellowship in Economics: \$13,325
- Faculty of Arts Graduate Award UBC: \$11,775
- International Tuition Award UBC: \$5,332
- President's Academic Excellence Initiative PhD Award UBC: \$652

2019

- Faculty of Arts Graduate Award UBC: \$25,524
- International Tuition Award UBC: \$2,133

2016

- **Summa Cum Laude in Law**: This honorific distinction is given to the students that have a GPA that is in the top 1% of his department in the last 5 years and display comprehensive achievements in an interview with the Academic Council
- **Cum Laude in Economics**: This honorific distinction is given to the students that have a GPA that is in

the top 5% of his department on the last 5 years

- [Commencement Speech at the graduation of Law, Architecture, Design and Humanities](#)
- [Commencement Speech at the graduation of Economics, Bussiness and Government](#)

2015

- Winner of the Jorge Méndez Munévar 2015 award: Annually given by the Department of Economics of the University of Los Andes to the student with the best research proyect in all the subjects of the B.A in Economics. This award was given due to the work *An Optimal Taylor Rule for Colombia: 2003-2015*
- Best National Higher Education Exam (Ecaes) in Law
- Winner of the Cavelier Association Competition and Scholarship First Place in the competition for the best article of investigation in Law, second semester of 2015, with the article "*The city, the property and the environment: Colective construction of a sustainable urban space*". As a consequence, I won a scholarship for eighth million colombian pesos

Students Advised

Master

- Tyler Wheeler - Laughlin Air Force Base
- Wyatt McHale - JD Berkeley Law

Programming

- Julia, Stata, Matlab, R, Arcgis

Languages

- English (fluent), Spanish (native), German (intermediate)